Advanced Statistical Mechanics
Assignment 01

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Question 1 (Yang-Lee edge):

Consider the following grand partition function for a hypothetical fluid

$$\Xi(z, V) = (1+z)^{[\alpha V]}(1+z^{[\alpha V]}).$$

Here $[x]$ denotes the integer part of x.

(a) Determine the zeros of $\Xi(z, V)$ and (roughly) plot them on the complex z plane. Here z denotes the fugacity, $z = e^{\beta\mu}$.

(b) Show that the zeros of $\Xi(z, V)$ in the thermodynamic limit hit the real z axis at $z = 1$.

(c) Evaluate

$$\Theta = \lim_{V\to\infty} \frac{1}{V} \ln \Xi(z, V)$$

for $z < 1$ and $z > 1$ and show that it is continuous but non-analytic.

(d) Given that the particle density is $n = z\frac{\partial\Theta}{\partial z}$, show that there is a density jump at the phase transition.

(e) Using the fact $\beta p = \Theta$, derive the equations of state $p(n)$, for $z < 1$ and $z > 1$ by eliminating z from the expressions of $p(z)$ and $n(z)$.

(a) To find the roots of Ξ, we set

$$(z+1)^{[\alpha V]} = 0 \qquad z^{[\alpha V]} = -1 \tag{1}$$

The real root is $z = -1$ with multiplicity $[\alpha V]$, which is physically irrelevant since the fugacity is $z = e^{\beta\mu} > 0$. The other complex roots come from $z^{[\alpha V]} = -1$, which are the roots of -1, lying on the unit circle

$$z_m = e^{i\theta_m} \qquad \theta_m = \frac{(2m+1)\pi}{[\alpha V]},\ m \in \{0, 1 \ldots [\alpha V]-1\}$$

(b) We see that the number of roots increases as we increase V. Consider the root with $m = 0$ and $n = [\alpha V] - 1$:

$$z_0 = e^{i\pi/[\alpha V]} \qquad z_1 = e^{i\pi(2[\alpha V]-1)/[\alpha V]}$$

(c) From the definition,

$$\begin{aligned}\Theta &= \lim_{V\to\infty} \frac{1}{V} \ln \Xi(z, V) \qquad z \in \mathbb{R}_{\geq 0} \\ &= \lim_{V\to\infty} \frac{1}{V} \left\{[\alpha V]\ln(1+z) + \ln\left(1 + z^{[\alpha V]}\right)\right\}\end{aligned} \tag{2}$$

Consider the second term.

Question 2 (Critical exponents in the mean field Ising model.):

Familiarize yourself with the definition of the critical exponents β, γ, and δ. Show that the specific heat for the mean field Ising model satisfies

$$\begin{cases} c = \dfrac{3Nk_B}{2}, & T < T_c, \\ c = 0, & T > T_c. \end{cases}$$

If the specific heat critical exponent α is defined via the relation

$$c \sim \left| \frac{T - T_c}{T_c} \right|^{-\alpha},$$

what value of α do we infer from the above analysis?

Answer. The partition function of the mean-field Ising model can be written as,

$$\mathcal{Z} = [2\cosh(\beta J\gamma m + \beta h)]^N e^{-\beta\varepsilon_0} \qquad \varepsilon_0 = \frac{JN\gamma m^2}{2}$$

Now, let us find the internal energy $U = -\partial_\beta \ln \mathcal{Z}$ for the zero-field case,

$$\begin{aligned} U &= -\frac{\partial}{\partial\beta}\Big(-\frac{\beta JN\gamma m^2}{2} + N\ln 2 + N\ln\cosh(\beta J\gamma m)\Big) \\ &= N\Big[\frac{J\gamma m^2}{2} + \beta J\gamma m\frac{\partial m}{\partial\beta} - \tanh(\beta J\gamma m)\Big(J\gamma m + \beta J\gamma\frac{\partial m}{\partial\beta}\Big)\Big] \end{aligned} \tag{3}$$

Using the self-consistent equation $m = \tanh(\beta J\gamma m)$ we get,

$$\begin{aligned} U &= N\Big[\frac{J\gamma m^2}{2} + \cancel{\beta J\gamma m\frac{\partial m}{\partial\beta}} - J\gamma m^2 - \cancel{\beta J\gamma m\frac{\partial m}{\partial\beta}}\Big] \\ &= -\frac{JN\gamma m^2}{2} \end{aligned} \tag{4}$$

The specific heat is defined as $c = \frac{\partial U}{\partial T}$ which can be written as,

$$c = \frac{\partial U}{\partial T} = \frac{\partial U}{\partial\beta} \times \frac{\partial\beta}{\partial T} = -k_{\mathrm{B}}\beta^2\frac{\partial U}{\partial\beta} = k_{\mathrm{B}}\beta^2\frac{JN\gamma}{2} \times 2m \times \frac{\partial m}{\partial\beta} \tag{5}$$

Now, consider the reduced emperature defined as $t = \frac{T_c - T}{T_c}$ where $k_{\mathrm{B}}T_c = J\gamma$,

$$\frac{1}{1-t} = \frac{T_c}{T} = \frac{J\gamma}{k_{\mathrm{B}}T} = \beta J\gamma \tag{6}$$

Hence the self-consistent equation becomes $m = \tanh\left(\frac{m}{1-t}\right)$ which can be expanded as,

$$m = \frac{m}{1-t} - \frac{m^3}{3(1-t)^3} + \ldots \tag{7}$$

from which we get either $m = 0$ or $m^2 = (3t)(1-t)^2$. For $T > T_c$, we have only one solution $m = 0$ and hence $\frac{\partial m}{\partial\beta} = 0$ which implies that

$$c = 0 \qquad \text{for} T > T_c \tag{8}$$

For $T < T_c$ we have,

$$m^2 = 3t + \mathcal{O}(t^2) \implies 2m \times \frac{\partial m}{\partial\beta} = 3\frac{\partial t}{\partial\beta} = 3\frac{\partial}{\partial\beta}\Big(1 - \frac{1}{\beta k_{\mathrm{B}}T_c}\Big) = \frac{3}{k_{\mathrm{B}}T_c\beta^2}$$

Substituting this in Eq. (5) we get for $T < T_c$,

$$c = \cancel{k_B\beta^2}\frac{JN\gamma}{2} \times \frac{3}{\cancel{k_B\beta^2}T_c} = \frac{3N}{2} \times \frac{J\gamma}{T_c} = \frac{3Nk_B}{2} \tag{9}$$

Thus, we obtain

$$c = \begin{cases} \frac{3Nk_B}{2} & T < T_c \\ 0 & T > T_c \end{cases} \tag{10}$$

Hence there is a discontinuous jump in specific heat across T_c. If we define the scaling

$$c \sim \left|\frac{T_c - T}{T_c}\right|^{-\alpha}$$

Then from the definition of the critical exponent, we have

$$\alpha = -\lim_{t\to 0^+} \frac{\ln\left(3Nk_B/2\right)}{\ln t} \to 0 \tag{11}$$

Hence, from the mean-field analysis, we obtain the critical exponent $\alpha = 0$.

Question 3 (q-state Potts model):

A generalization of the Ising model is the q-state Potts model, where one assume that the spin variable at a site can now take q different values, $\sigma_i \in \{1, 2, 3, .., q\}$ and the Hamiltonian is given by,

$$\mathcal{H}_{\text{Potts}} = -J\sum_{\langle ij\rangle}\delta_{\sigma_i\sigma_j} - h\sum_i \delta_{\sigma_i,1}$$

where $\langle ij\rangle$ denotes nearest neighbour interaction. By defining a suitable transformation between the Ising spin variable σ_i and the Kronecker delta $\delta_{\sigma_i\sigma_j}$, show that the for $q = 2$, this Hamiltonian is equivalent to the Ising Hamiltonian.

Answer. The interaction term given here is $\delta_{\sigma_i\sigma_j}$ while in the Ising Hamiltonian, the interaction term is S_iS_j. If both Ising spins are different, the interaction is -1 while for the $q = 2$ Potts model, it is 0. If both Ising spins are same, the interaction is $+1$ which is same as in the $q = 2$ Potts model. Thus we can have a linearly transformation, such that

$$\delta_{\sigma_i\sigma_j} = aS_iS_j + b$$

The conditions to satisfy are,

$$0 = b - a \qquad 1 = b + a$$

from which we obtain $a = b = 1/2$. Thus, the transformation is,

$$\delta_{\sigma_i\sigma_j} = \frac{1}{2}(S_iS_j + 1)$$

Essentially we are mapping $S_i = +1 \mapsto \sigma_i = 1$ and $S_i = -1 \mapsto \sigma_i = 2$. Thus the second Kronecker delta becomes,

$$\delta_{\sigma_i,1} = \frac{1}{2}(S_i + 1)$$

With this transformation, the Potts Hamiltonian becomes,

$$\begin{aligned}
\mathcal{H}_{\text{Potts}} &= -\frac{J}{2}\sum_{\langle ij\rangle}(S_iS_j + 1) - \frac{h}{2}\sum_i(S_i + 1) \\
&= -\frac{J}{2}\sum_{\langle ij\rangle}S_iS_j - \frac{h}{2}\sum_i S_i - \underbrace{\frac{J}{2}\sum_{\langle ij\rangle}1 - \frac{h}{2}\sum_i 1}_{\text{const.}} \\
&= -J'\sum_{\langle ij\rangle}S_iS_j - h'\sum_i S_i + \text{const.} \\
&= \mathcal{H}_{\text{Ising}} + \text{const.}
\end{aligned} \tag{12}$$

where $J' \equiv {}^J\!/_2$ and $h' \equiv {}^h\!/_2$. Since a constant in the Hamiltonian can be neglected (will cause only a shift in the energy), we obtain that for $q = 2$, the Potts model reduces to the Ising Hamiltonian.

We define the following transition in the spins.

Question 4 (Antiferromagnetic Ising Model):

Consider the Ising model on a two-dimensional square lattice with nearest neighbour interaction but with the interaction term having opposite sign such that antiparallel allignment is favoured.

$$\mathcal{H} = J\sum_{\langle ij\rangle}\sigma_i\sigma_j - h\sum_i \sigma_i \qquad J > 0$$

(a) Show that for $h = 0$, the partition function and hence the free energy is the same as that for the usual Ising model.

(b) Solve the problem in the mean field approximation with $h = 0$.

Answer. (a) Let us divide the 2D lattice into two sub-lattices with sites A and site B, alternately placed. Hence, there are only sites of type B surrounding any site of type A (and vice versa). Let us consider the zero-field Ising model on this sublattice.

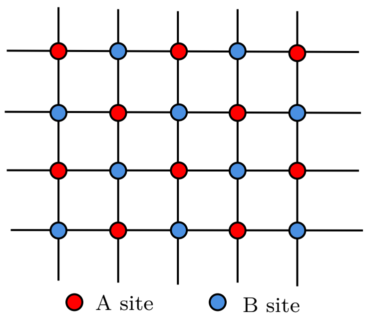

Figure 1: 2D lattice broken into interpenetrating sub-lattices.

Since interaction of spins on type A site is only with those at type B site, the Ising Hamiltonian $\mathcal{H}(h = 0, J, \{\sigma_i\}) = -J\sum_{\langle ij\rangle}\sigma_i\sigma_j$ can be written as,

$$\mathcal{H}(h = 0, J, \{\sigma_i^A\}, \{\sigma_i^B\}) = -J\sum_{\langle ij\rangle}\sigma_i^A\sigma_j^B \tag{13}$$

From this, the obvious symmetry exhibited by the Hamiltonian is,

$$\mathcal{H}(h = 0, -J, \{\sigma_i^A\}, \{\sigma_i^B\}) = \mathcal{H}(h = 0, J, \{-\sigma_i^A\}, \{\sigma_i^B\}) = \mathcal{H}(h = 0, J, \{\sigma_i^A\}, \{-\sigma_i^B\}) \tag{14}$$

The zero-field partition function becomes,

$$\begin{aligned}\mathcal{Z}(h = 0, -J, T) &= \sum_{\{\sigma^A\}}\sum_{\{\sigma^B\}} e^{-\beta\mathcal{H}(0,-J,\{\sigma^A\},\{\sigma^B\})}\\ &= \sum_{\{\sigma^A\}}\sum_{\{\sigma^B\}} e^{-\beta\mathcal{H}(0,J,\{-\sigma^A\},\{\sigma^B\})}\\ &= \sum_{\{\sigma^A\}}\sum_{\{\sigma^B\}} e^{-\beta\mathcal{H}(0,J,\{\sigma^A\},\{\sigma^B\})}\\ &= \mathcal{Z}(h = 0, J, T)\end{aligned} \tag{15}$$

Going from the first to the second line, we used the sub-lattice symmetry from Eq. (14) while going from the second to the third line, we used the fact the since we are summing over all values of $\sigma_i = \pm$, the sum with $\{-\sigma\}$ will be the same as the sum with $\{+\sigma\}$. Since the free energy is just the derivative of the partition function, we obtain

$$F(h=0,J,T) = F(h=0,-J,T) \tag{16}$$

Hence the partition function and the free energy for the ferromagnetic ($J>0$) and the anti-ferromagnetic ($J<0$) Ising model is same and hence, both these are thermodynamically equivalent.

(b) We apply the mean-field approximation where we neglect the fluctuations in the spin.

$$(m_{\mathrm{A}} - \sigma_i^A)(m_{\mathrm{B}} - \sigma_j^B) = m_{\mathrm{A}} m_{\mathrm{B}} - m_{\mathrm{A}}\sigma_j^B - m_{\mathrm{B}}\sigma_i^A + \sigma_i^A \sigma_j^B = 0 \tag{17}$$

Substituting this in the anti-ferromagnetic Hamiltonian with $J>0$ we get,

$$\begin{aligned}\mathcal{H} = J\sum_{\langle ij\rangle}\sigma_i^A\sigma_j^B &= Jm_{\mathrm{A}}\sum_{\langle ij\rangle}\sigma_j^B + Jm_{\mathrm{B}}\sum_{\langle ij\rangle}\sigma_j^A - Jm_{\mathrm{A}}m_{\mathrm{B}}\sum_{\langle ij\rangle}1 - h\sum_i\sigma_i^A - h\sum_j\sigma_j^B \\ &= -(h - Jm_{\mathrm{A}}\gamma)\sum_i\sigma_i^B - (h - Jm_{\mathrm{B}}\gamma)\sum_i\sigma_i^A - \frac{Jm_{\mathrm{A}}m_{\mathrm{B}}\gamma}{2} \\ &= -h_{\mathrm{A}}\sum_i\sigma_i^B - h_{\mathrm{B}}\sum_i\sigma_i^A - \frac{Jm_{\mathrm{A}}m_{\mathrm{B}}\gamma}{2}\end{aligned}$$

The partition function becomes,

$$\mathcal{Z} = \mathcal{Z}_{\mathrm{A}}^{N/2}\mathcal{Z}_{\mathrm{B}}^{N/2}e^{-\beta J\gamma m_{\mathrm{A}}m_{\mathrm{B}}/2}$$

where $\mathcal{Z}_{\mathrm{A}} = 2\cosh(\beta h_{\mathrm{B}})$ and $\mathcal{Z}_{\mathrm{B}} = 2\cosh(\beta h_{\mathrm{A}})$. The sublattice magnetisation can be obtained from the partition function as,

$$m_{\mathrm{A}} = \tanh(\beta h_{\mathrm{B}}) \qquad m_{\mathrm{B}} = \tanh(\beta h_{\mathrm{A}}) \tag{18}$$

which are two coupled transcendental equations. In the zero field, $h_{\mathrm{A}} = -Jm_{\mathrm{A}}\gamma$ and $h_{\mathrm{B}} = -Jm_{\mathrm{B}}\gamma$. Expanding $\tanh(x)$ for $x \ll 1$ we get,

$$\begin{aligned}m_{\mathrm{A}} &= -\tanh\left(m_{\mathrm{B}}\frac{T_{\mathrm{c}}}{T}\right) \simeq -\frac{T_{\mathrm{c}}}{T}m_{\mathrm{B}} + \frac{1}{3}\left(\frac{T_{\mathrm{c}}}{T}\right)^3 m_{\mathrm{B}}^3 \\ m_{\mathrm{B}} &= -\tanh\left(m_{\mathrm{A}}\frac{T_{\mathrm{c}}}{T}\right) \simeq -\frac{T_{\mathrm{c}}}{T}m_{\mathrm{A}} + \frac{1}{3}\left(\frac{T_{\mathrm{c}}}{T}\right)^3 m_{\mathrm{A}}^3\end{aligned} \tag{19}$$

Define $M := \frac{m_{\mathrm{A}}+m_{\mathrm{B}}}{2}$ and $m_{\mathrm{s}} := \frac{m_{\mathrm{A}}-m_{\mathrm{B}}}{2}$ which allows us to write,

$$\begin{aligned}m_{\mathrm{A}}^3 + m_{\mathrm{B}}^3 &= 2M(M^2 + 3m_{\mathrm{s}}^2) \\ m_{\mathrm{A}}^3 - m_{\mathrm{B}}^3 &= 2m_{\mathrm{s}}(3M^2 + m_{\mathrm{s}}^2)\end{aligned} \tag{20}$$

From Eq. (19), by adding and subtracting we have:

$$\begin{aligned}M\left(1 + \frac{T_{\mathrm{c}}}{T}\right) &= \frac{M}{3}\left(\frac{T_{\mathrm{c}}}{T}\right)^3\left[M^2 + 3m_{\mathrm{s}}^2\right] \\ m_{\mathrm{s}}\left(1 - \frac{T_{\mathrm{c}}}{T}\right) &= -\frac{m_{\mathrm{s}}}{3}\left(\frac{T_{\mathrm{c}}}{T}\right)^3\left[3M^2 + m_{\mathrm{s}}^2\right]\end{aligned} \tag{21}$$

The solution of the second equation for is $m_{\mathrm{s}} = 0$ or h